

Academic Programs

PEOs, POs & PSOs

B.Tech : CSE - Data Science

Program Educational Objectives (PEOs)

The student in CSE - Data Science will be able to:

Graduates of the Data Science will contribute to the

Nation's progress through their talent to solve varied and complex Data Science problems across a broad range of application areas. (Professional accomplishment).

Graduates of the Data Science will be successful

Professionals, designing and implementing Products and services of global standards in the field Data Science, becoming entrepreneurs, pursuing higher studies and research. (Continuing Education).

Graduates of the Data Science Program will be able to adapt

to changing scenario of disruptive technology with an ability to solve critical societal problems using logical and flexible approach in decision making. (Attitudes).

To make the graduates adaptable to changing career

opportunities who have the potential to excel in industry or public sector or higher studies exhibiting global competitiveness.

Program Specific Outcomes (PSOs)

Acquire proficiency in developing and implementing efficient

solutions using evolving technologies of Data Science, platforms and Free and Open-Source Software.

Gain critical understanding of hardware and software tools

providing to the contemporary needs of IT industry 4.0 standards.

Graduating student will be able to illustrate the use of

Data Science for attaining reliable and fault-tolerant IT solutions by using and understanding use cases.

B.Tech : CSE- Cyber Security

The student in CSE- Cyber Security will be able to:

Provide sustainable solutions for real time cyber-attacks as

a part of professional career and research. (Professional accomplishment).

Practice lifelong learning through pursuing higher

education, professional programs, training and certifications. (Continuing Education).

Demonstrate proactive team spirit, competencies, and ethical

behaviour in their careers. (Attitudes).

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Generate innovative solutions and explore opportunities to become entrepreneurs.

Explore, employ, and support security systems for the networked devices and data.

Respond to security events, and protect devices and systems from cybersecurity risks, threats, and vulnerabilities.

Provide awareness to people regarding the cyber security threats and propagate ethical standards.

B.Tech : Artificial Intelligence & Data Science

The student in Artificial Intelligence & Data Science will be able to:

Enable the students to accomplish professional career in business, government, or academia, with the ability to create creative solutions utilising technology as a tool to address pressing issues. (Professional accomplishment).

Conduct research talents in cutting-edge technologies will add to a new corpus of knowledge (Continuing Education).

Develop a learning mind-set to continuously improve their knowledge, through on the job, formal and informal learning opportunities. (Attitudes).

Build ethical ability and good leadership, management, teamwork, and communication abilities.

Apply the principles of artificial intelligence and data science that require problem-solving, inference, perception, knowledge representation, and learning.

Exhibit strong professional skills to function effectively in multi-disciplinary and heterogeneous teams with a growth mind-set to deliver a quality product for Business, Education & Training and E-governance.

Provide a concrete foundation and enrich their abilities to qualify for Employment, Higher studies and Research in Artificial Intelligence and Data science with ethical values.

Program Outcomes (POs):

The graduates of B. Tech CSE-(CyS, DS) and AI&DS will be able to:

Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions

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using first principles of mathematics, natural sciences, and engineering sciences.

Design/development of solutions: Design

solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

Conduct investigations of complex problems:

Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions for complex problems.

Modern tool usage: Create, select, and

apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

The engineer and society: Apply reasoning

informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

Environment and sustainability:

Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

Ethics: Apply ethical principles and

commit to professional ethics and responsibilities and norms of the engineering practice.

Individual and team work: Function

effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

Communication: Communicate effectively on

complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

Project management and finance:

Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects

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and in multidisciplinary environments.

Life-long learning: Recognize the need

for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.